Journal of NeuroEngineering and Rehabilitation

RESEARCH

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Monitoring wheelchair propulsion patterns: feasibility and validity of using wearable sensors

Ramin Fathian[1], Aminreza Khandan[1], Nasim Rahmanifar[1^], Chester Ho[2,3] and Hossein Rouhani[1,3*]

Abstract

Currently, there is a need to understand the characteristics of manual wheelchair propulsion patterns in the daily life of users and the impact of these patterns on repetitive strain injury of the shoulders. This study aimed to develop a method for remote and long-term monitoring of wheelchair propulsion techniques. We used a hand-mounted inertial measurement unit (IMU) to identify propulsion patterns in manual wheelchair users. IMU data was collected from 12 participants (7 males and 5 females), including 8 experienced and 4 inexperienced manual wheelchair users. We applied continuous wavelet transform (CWT) for feature extraction and used Support Vector Machine (SVM) and Multilayer Perceptron (MLP) Neural Network for pattern classification. SVM with a linear kernel achieved 89% accuracy, 78% F1-score, 78% precision, and 78% recall. SVM with a polynomial kernel achieved 94% accuracy, 88% F1-score, 88% precision, and 89% recall, while the MLP reached 95% accuracy, 89% F1-score, 89% precision, and 89% recall. Neither the participants' wheelchair experience nor their gender significantly affected the performance of the classifiers. These findings suggest that the proposed IMU and propulsion patterns classification method can be used across different user profiles for remote and long-term monitoring of wheelchair propulsion patterns to better understand shoulder overuse risk in daily life.

Keywords Remote monitoring, Manual wheelchair, Propulsion pattern, Wearable technology, Shoulder injury

Introduction

Manual wheelchairs are the primary means of ambulation for many individuals with mobility impairments due to fatigue, weakness, or balance issues as a result of aging, cardiovascular, respiratory, and neuromuscular problems. It is estimated that around 1% and at least 2% of the population in developed and developing countries are manual wheelchair users [4, 27, 34] Worldwide Need - Wheelchair Foundation [37]. The repetitive motion involved in propelling a manual wheelchair places significant strain on the upper limb and has been hypothesized to be a potential cause of repetitive strain injury of the upper extremities. Previous studies have identified upper-limb soft-tissue pathology (e.g., muscle/tendon injury, bursopathy), along with degenerative joint changes and peripheral neuropathies, as demonstrated by imaging and neurophysiologic evaluation [2, 5, 20, 26, 29, 32, 35]. Adopting effective propulsion methods has been proven to decrease shoulder load, thereby reducing the prevalence of pain and complications in the upper extremities [3, 24, 31, 36].

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Clinical practice guidelines suggest that the semicircular propulsion pattern (Fig. 1) and an extended push duration are linked to reduced upper extremity pain and strain injuries due to overuse (Consortium for Spinal Cord Medicine. & Paralyzed Veterans of America., [7]). Simulation studies revealed that the semicircular and double-loop patterns require the least muscle effort compared to the single-loop and arc patterns [33]. Additionally, the single-loop pattern is associated with higher cadence, lower push percentage, and smaller contact angle compared to semicircular and double-loop patterns [2, 22]. These factors—cadence, push-to-recovery percentage, and contact angle—are considered risk factors for shoulder pain and could affect the prevalence of shoulder injury [29, 32]. Despite the recognized connection between manual wheelchair activities and shoulder pain, there is a lack of methods to monitor and evaluate the cumulative exposure to repetitive upper limb tasks experienced by manual wheelchair users. Prior studies utilized wheel-mounted or body-worn sensors to monitor the physical activity of these users [9, 12, 13, 15, 21, 25, 28]. Earlier, French et al. [12] demonstrated feasibility of using wrist-mounted two-axis accelerometer to classify of the four classic propulsion patterns in a single-subject, multi-surface study. Goodwin et al. [13] and Kressler et al. [21] suggested accelerometer-based methods to estimate arm usage and activities. Similarly, Leving et al. [25] and Popp et al. [28] utilized commercial IMUs to categorize wheelchair-related activities and determine energy expenditure for persons with spinal cord injury, respectively. Hiremath et al. [14, 15] introduced a gyroscope and accelerometer-based system to assess physical activity. Prior IMU-based studies mostly focused on activity-level parameters such as propulsion time vs. non-propulsion time, propulsion mode (self-propelled vs. attendant-pushed), distance and speed, and energy-expenditure estimates [13–15,18, 19, 25, 28]. These measures do not provide cycle-level details such as the propulsion pattern, push time, or recovery time. It should be mentioned that, IMUs could be leveraged to estimate cycle-level kinematics (propulsion pattern, cadence, push and recovery durations); however, quantifying mechanical load (push-rim forces, joint moments)—particularly during high-demand tasks in which external load is elevated—requires kinetic measurements [8, 20, 23, 30, 33]. This gap limits our ability to quantify cumulative exposure to stroke-level risk factors in daily use. Our study addresses this gap by using a hand-mounted IMU and a CWT-based pipeline to classify the four common propulsion patterns across participants, and by outlining cycle-level markers (e.g., push–recovery proportion, cadence, pattern prevalence) that could serve as practical proxies for exposure in remote, long-term monitoring.

To address these concerns—limited understanding of propulsion characteristics that may reduce upper-extremity pain and strain—the following objectives were studied: (1) Characterize the four common propulsion patterns with a hand-worn IMU and suggest classification features; (2) Assess preliminary feasibility of propulsion patterns monitoring by estimating cross-validated pattern-classification performance. This will pave the way toward having a detailed estimation of cumulative exposure to repetitive upper limb tasks during manual wheelchair propulsion.

Methods

Study overview

To investigate the feasibility and validity of using wearable sensors to classify propulsion patterns, we collected experimental data from a hand-mounted IMU and developed a two-stage pipeline. In the first stage, we identified the manual wheelchair propulsion cycles [10]; in the second stage, we classified the propulsion pattern employed by the user. To evaluate our method we focused on: (i) generalizability, defined as subject-independent performance using one-participant-out cross-validation; (ii) robustness, defined as stable performance across one-participant-out folds and propulsion patterns

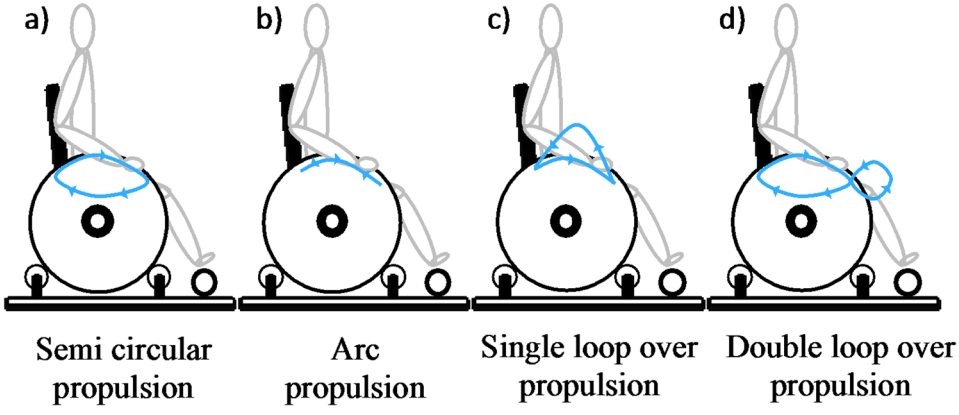


Fig. 1 Four common wheelchair propulsion patterns: **a** semicircular, **b** arc, **c** single-loop over, and **d** double-loop over

(per-pattern accuracy and F1-score); and (iii) adaptability, defined as tolerance to variability in cadence and cycle duration achieved by normalizing each detected push cycle to a fixed-length representation and applying a single, fixed pipeline across participants, without per-subject tuning. So, we investigated the performance of our method using accuracy, F1-score, precision, and recall in identifying the propulsion patterns using the leave-one-subject-out methodology. Additionally, we contextualized our results against published state-of-the-art data-driven approaches where available.

Wheelchair propulsion pattern

There are four common patterns, semicircular (SC), arc (Arc), single-loop over (SL), and double-loop over (DL) (Fig. 1), that could also be categorized into under the rim (SC and Arc) and over the rim patterns (SL and DL) propulsion.

Experimental design, setup, and procedure

A manual wheelchair was placed on a roller platform to allow participants to propel the wheelchair freely in the lab environment (Fig. 2). Twelve participants were recruited for the study through institutional posting at University of Alberta (Edmonton, AB, Canada), Glenrose Rehabilitation Hospital (Edmonton, AB, Canada) and Spinal Cord Injury Alberta (Alberta, Canada). The cohort comprised both experienced ($n=8$) and inexperienced ($n=4$) manual wheelchair users. The population included 7 males and 5 females, with a mean age of 34 ± 8 years, a mean height of 174 ± 11 cm, and a mean weight of 65 ± 11 kg. The participants' conditions were as follows: five individuals with paraplegia, two individuals with spina bifida, one individual with cerebral palsy, and four non-disabled individuals. Experienced group consisted of daily wheelchair users, while the inexperienced group included able-bodied individuals, representing those new to wheelchair use. An IMU (MTws, XSENS Technologies, NL) with a sampling rate of 100 Hz was placed on the participants' dominant hand using double-sided medical tape to record the hand motion while propelling the wheelchair (Fig. 2).

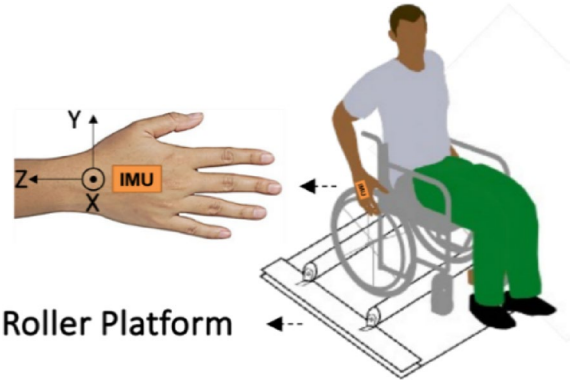


Fig. 2 Experiment setup, including the wheelchair platform and the IMU sensor. The IMU placement on the hand and associated recordings axis are illustrated

Participants were instructed to propel the wheelchair on the roller platform and familiarize themselves with the instrumented wheelchair and roller platform setup. We used the platform in this feasibility phase to standardize the overall methodology, obtain balanced and clearly labeled trials for each propulsion pattern, and reduce confounders (e.g., surface friction, path curvature, turning). While the roller platform enables consistent, repeatable propulsion data collection, it inherently restricts movement to straight-line propulsion, which is a known limitation of such laboratory setups. This constraint, while suitable for controlled data collection, does not replicate the full range of propulsion conditions encountered in daily wheelchair use, such as turning or obstacle negotiation. During each data collection session, participants were asked to propel the wheelchair employing one of four common wheelchair propulsion patterns: SC, Arc, SL, and DL, for one minute each. The experimental protocol was approved by the University of Alberta's Research Ethics Board Committee (No. Pro00115722). To identify the propulsion cycles, peaks in the IMU measurements that correspond to the start and end of a propulsion cycle were detected using a custom-developed algorithm [10] which was validated previously, achieving 96% event detection accuracy, with average F1-score 98%, precision 96%, and sensitivity 99%. This ensured accurate cycle detection regardless of the propulsion pattern employed by the participant.

Propulsion pattern classification

IMU measurements were filtered using a no-lag 6th-order low-pass Butterworth filter with a cut-off frequency of 10 Hz to remove high-frequency noise and minor vibration artefacts at the skin–sensor interface. This filtered signal was used only to detect individual propulsion cycles more reliably. Then, the unfiltered IMU measurements were segmented and stored separately based on the identified propulsion cycles [10]. Each trial instructed a single propulsion pattern (SC, Arc, SL, or DL) and were labeled accordingly. To avoid bias and class imbalance, 40 random recordings for each of the four patterns from each participant (160 in total from each participant) were selected. Classification performance and error analyses were obtained using leave-one-subject-out cross-validation: in each fold, one participant (160 recordings) was held out for testing and the model was trained on the remaining participants; metrics were computed on the held-out participant and averaged across folds. To characterize the propulsion technique using IMU measurements, first, we used the continuous wavelet transform

(CWT) to obtain a time-frequency representation of the IMU measurements during the propulsion. This method was chosen because propulsion cycles are non-stationary and their spectral content changes within each cycle. The Morlet wavelet was selected as the mother wavelet. The CWT was computed over a range of scales from 1 to 450. In the context of the Morlet wavelet, these scale values correspond to frequency components approximately spanning the 0.1 Hz to 100 Hz range. We selected this range to ensure that the spectral characteristics of both low-frequency components (general hand motion) and high-frequency components (brief, pattern-specific transients at push-rim contact onset/offset) were adequately captured. We extracted spectral characteristics representing manual wheelchair propulsion. To enable consistent comparison across propulsion cycles of varying durations, the resulting CWT coefficient matrix was temporally resampled to a fixed length of 2000 samples using Piecewise Cubic Hermite Interpolating Polynomial. This resampling was applied only to the time axis of the CWT matrix, not the original signal, preserving the integrity of spectral content while standardizing input length for segmentation and feature extraction. Each normalized propulsion cycle was divided into five equal-length temporal windows (400 samples each). This windowing strategy was applied to retain localized temporal structure within each cycle. By extracting features from these windows separately, we preserved time-resolved information that supported the differentiation of propulsion patterns with varying dynamics (Fig. 3). From the resulting 2250 (5 × 450) values, 6 features, including Kurtosis (1) (measure that quantifies the shape of a distribution), Root Mean Square (2), Maximum Value (3), Crest Factor (measure the ratio of the peak value to the root mean square) (4), Shape Factor (dimensionless value that characterizes the shape of a signal providing information about the symmetry or asymmetry of data) (5) and Impulse Factor (identifying sudden and high amplitude spikes in the signal) (6), were computed in time and scale domains [17]. These features were selected to summarize distribution shape, signal energy, and peak-to-average behavior across time–scale bins; these descriptors are commonly used for non-stationary, vibration-like signals and were chosen for interpretability and robustness in a modest-sample setting [17]. The x denotes the individual data points that the features were extracted from (Fig. 3). i = 1:450 denotes the scale index and m = 1:400 denotes the within-segment sample index. Individual coefficients were written as $x_{i,m}$, $\bar{x}_m$ represented the average (mean) value of the mth values across all i, and σ_m represented the standard deviation (7).

$$F_{Kurtosis,\ m} = \frac{\sum_{i=1}^{n}(x_{i,m} - \bar{x}_m)^4}{(n-1)\sigma_m^4} \quad (1)$$

$$F_{Root\ Mean\ Square,\ m} = \left(\frac{1}{n}\sum_{i=1}^{n} x_{i,m}{}^2\right)^{0.5} \quad (2)$$

$$F_{Max\ Value,\ m} = \max\left(x_{i,m}\right), i = 1\ to\ n \quad (3)$$

$$F_{Crest\ Factor,\ m} = \frac{\max\left(x_{i,m}\right)}{\left(\left(\frac{1}{n}\sum_{i=1}^{n} x_{i,m}^2\right)^{0.5}\right)},\ i = 1\ to\ n \quad (4)$$

$$F_{Shape\ Factor,m} = \frac{\left(\frac{1}{n}\sum_{i=1}^{n} x_{i,m}{}^2\right)^{0.5}}{\bar{x}_m} \quad (5)$$

$$F_{Impulse\ Factor,m} = \frac{\max\left(x_{i,m}\right)}{\bar{x}_m},\ i = 1\ to\ n \quad (6)$$

$$\sigma_m = \left(\frac{1}{n-1}\sum_{i=1}^{n}(x_{i,m} - \bar{x}_m)^2\right)^{0.5} \quad (7)$$

All these features (5 segments ×36 features per segment) from each propulsion cycle were then used to train the classifiers (Figs. 3 and 4).

This study included Support Vector Machines (SVM) and Multilayer Perceptron (MLP) Neural Network due to their demonstrated effectiveness on datasets with modest sample sizes [1, 16]. Two different kernel functions (linear and polynomial) were used for the SVM. The MLP included an input layer of 180 neurons (corresponding to the 180 input features), three hidden layers with 128, 64, and 32 neurons, respectively, and an output layer with 4 neurons (corresponding to the 4 classes). Rectified linear units and softmax activation functions were used for the three hidden layers and the output layer, respectively. Backpropagation was used to train the MLP for 200 epochs using Adam (learning rate 0.001), batch size 64, and cross-entropy loss. We used Xavier (Glorot) uniform initialization for the weights and set the random seed to 42 for reproducibility. We applied early stopping on validation loss (patience 20; best weights restored). Beyond choosing kernel type (linear vs. polynomial) and using standard, conservative settings for the MLP (learning rate 0.001, Adam, cross-entropy), we performed small sensitivity checks (e.g., varying SVM regularization strength and modestly changing MLP width), which did not alter the outer-fold results or conclusions. We intentionally avoided extensive hyperparameter tuning to reduce optimistic bias in a small-sample regime. Using these relatively simple classifiers allowed us to evaluate the representational power of the extracted CWT features and assess their feasibility in distinguishing propulsion

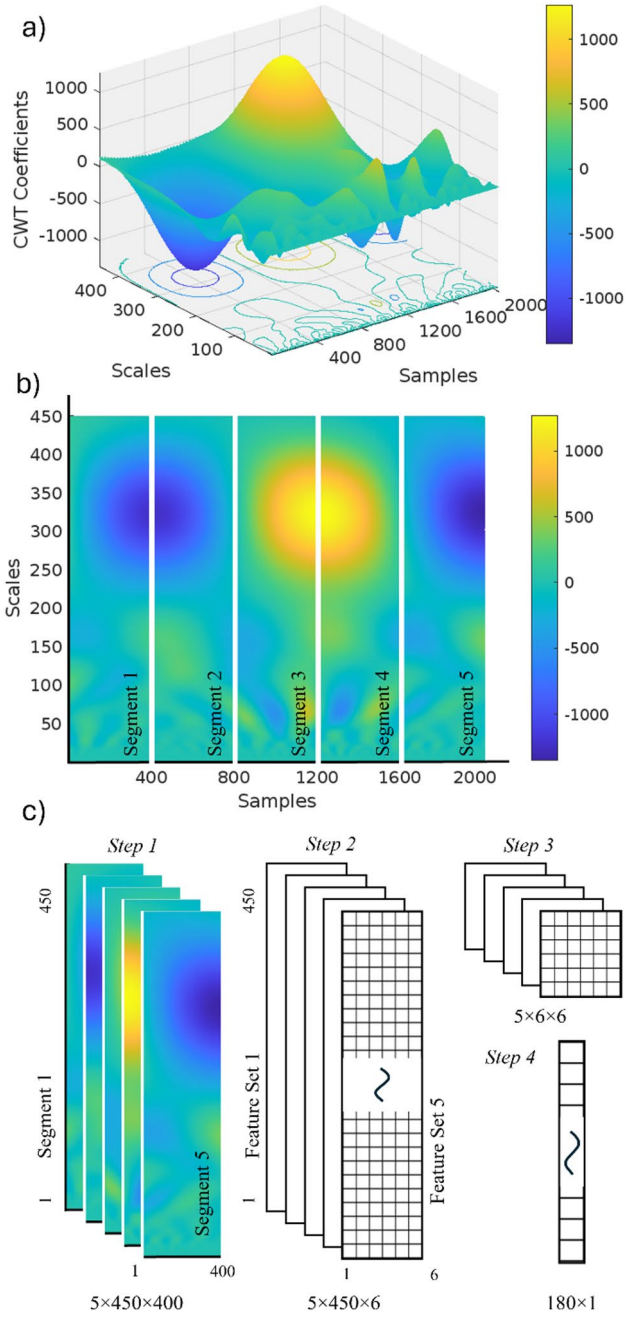


Fig. 3 The flow of our feature extraction method for identifying propulsion patterns. **a**: Time-frequency representation of a single normalized propulsion cycle obtained using continuous wavelet transform (CWT) with the Morlet wavelet. The x-axis corresponds to the normalized time (resampled to 2000 points), and the y-axis shows the frequency content across scales (1–450). **b**: CWT coefficients obtained during normalized propulsion cycle are divided into five equal lengths in the time domain to allow localized feature extraction across different cycle phases. **c**: Step 1 involves calculating six features—Kurtosis, Root Mean Square, Maximum Value, Crest Factor, Shape Factor, and Impulse Factor—for each of the five segments across all 450 scales along the temporal axis, resulting in a matrix of dimensions 5×450×6. In Step 2, these six features are recalculated, this time for each of the six time-domain features across the scales axis, resulting in a matrix of 5×6×6. Then, the entire set of features extracted from the propulsion cycle (Step 3) is converted into a flattened array of 180×1 (Step 4)

patterns. This decision was made to investigate the feasibility of using CWT for feature extraction in manual wheelchair propulsion pattern classification.

Performance analysis

Each classifier was evaluated using a one-participant-out cross-validation approach on the IMU recordings, resulting in a total of 12 folds. To assess classification performance and validity of the proposed method in identifying propulsion patterns, we calculated accuracy, F1-score, precision, and recall for each propulsion pattern and classifier. These metrics were averaged across all folds to identify overall trends and class-specific performance disparities. Additionally, we examined the potential influence of participant gender and wheelchair experience on classifier performance by applying a non-parametric Friedman test to the performance metrics. Standard deviations were also reported to characterize the consistency of classifier performance across folds.

Results

The performance of SVM and MLP classifiers in detecting different propulsion patterns was investigated using 12-fold one-participant-out cross-validation. The dataset was class-balanced (480 cycles per pattern; 160 per participant; 1920 cycles total), ensuring that performance metrics were not confounded by class imbalance.

SVM approach

Using SVM with linear and polynomial kernels, we achieved average accuracies of 89% and 94%, respectively. The F1-score of 78% and 88%, precision of 78% and 88%, and recall of 78% and 89% were achieved for each kernel, respectively. For each classifier, accuracy, F1-score, precision, and recall were computed per propulsion pattern in each fold of the 12-fold cross-validation, then averaged across folds (Tables 1 and 2). Our results indicated that the polynomial kernel demonstrated superior performance compared to the linear kernel, illustrating better classification accuracy, F1-score, precision, and recall (Tables 1 and 2).

MLP approach

The MLP achieved accuracy, F1-score, precision, and recall of 95%, 89%, 89%, and 89% after 200 epochs (Table 3). Performance metrics were calculated per propulsion pattern in each fold of the 12-fold cross-validation and then averaged across folds (Table 3). An average increase of 5% and 6% in accuracy, 10% and 11% in F1-score, 10% and 11% precision, 11% and 11% and recall were observed when comparing SVM with the polynomial kernel to SVM with linear kernel and comparing MLP to SVM with linear kernel, respectively (Table 4). However, we observed no significant difference

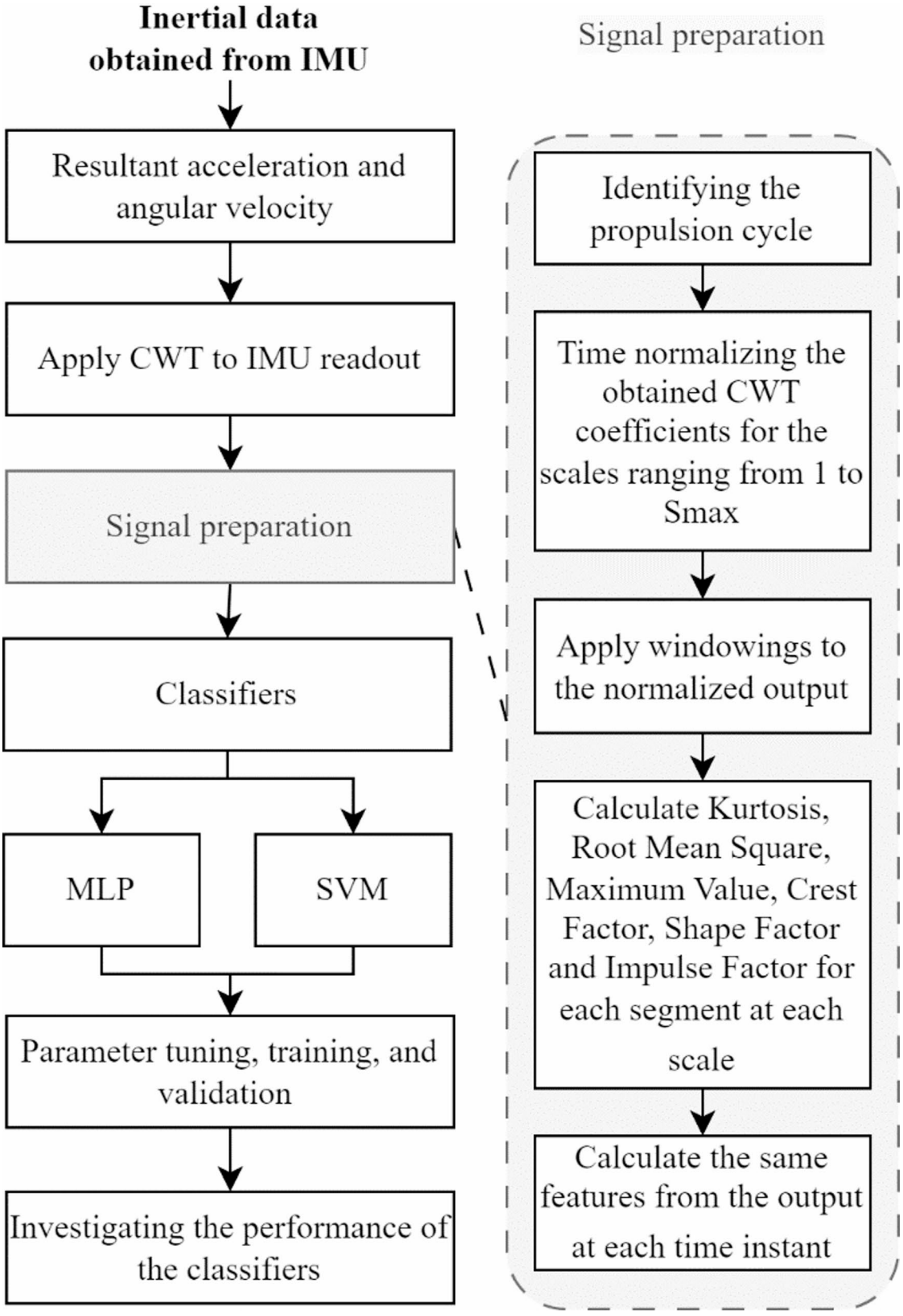


Fig. 4 Overview of the pipeline from IMU readout to classification and evaluation. After cycle identification and feature extraction, the data are passed to one classifier at a time (either MLP or SVM) for training and testing under a leave-one-subject-out cross-validation. Performance metrics are reported per model and averaged across folds

Table 1 Overall SVM (Linear) classification performance in detecting propulsion patterns

	Accuracy	F1-score	Precision	Recall
SC	**0.90 ± 0.02**	**0.79 ± 0.04**	0.79 ± 0.04	**0.79 ± 0.04**
Arc	0.88 ± 0.02	**0.79 ± 0.04**	**0.80 ± 0.04**	0.78 ± 0.04
SL	**0.90 ± 0.02**	0.75 ± 0.04	0.74 ± 0.05	0.77 ± 0.05
DL	0.88 ± 0.02	0.78 ± 0.05	0.79 ± 0.04	0.78 ± 0.04
Average	0.89 ± 0.02	0.78 ± 0.04	0.78 ± 0.04	0.78 ± 0.04

The best metrics for each propulsion pattern are shown in bold

Table 2 Overall SVM (Polynomial) classification performance in detecting propulsion patterns

	Accuracy	F1-score	Precision	Recall
SC	0.94 ± 0.02	0.88 ± 0.03	0.88 ± 0.03	0.88 ± 0.03
Arc	0.94 ± 0.03	**0.89 ± 0.03**	**0.90 ± 0.03**	0.88 ± 0.04
SL	**0.95 ± 0.02**	0.87 ± 0.04	0.86 ± 0.04	**0.89 ± 0.03**
DL	0.94 ± 0.02	**0.89 ± 0.03**	**0.9 ± 0.03**	**0.89 ± 0.03**
Average	0.94 ± 0.02	0.88 ± 0.03	0.88 ± 0.03	0.89 ± 0.03

The best metrics for each propulsion pattern are shown in bold

Table 3 Overall MLP classification performance in detecting propulsion patterns

	Accuracy	F1_Score	Precision	Recall
SC	**0.95 ± 0.02**	0.89 ± 0.03	0.90 ± 0.04	0.89 ± 0.04
Arc	0.94 ± 0.02	**0.90 ± 0.03**	0.89 ± 0.04	**0.91 ± 0.03**
SL	**0.95 ± 0.02**	0.87 ± 0.04	0.86 ± 0.05	0.89 ± 0.04
DL	0.94 ± 0.02	**0.90 ± 0.04**	**0.92 ± 0.03**	0.88 ± 0.05
Avg	0.95 ± 0.01	0.89 ± 0.03	0.89 ± 0.03	0.89 ± 0.03

The best metrics for each propulsion pattern are shown in bold

Table 4 Overall performance of classifiers in detecting propulsion patterns (presented in mean ± standard deviation)

	SVM (Linear)	SVM (Polynomial)	MLP
Accuracy	0.89 ± 0.02	0.94 ± 0.02	**0.95 ± 0.02**
F1-score	0.78 ± 0.04	0.88 ± 0.03	**0.89 ± 0.03**
Precision	0.78 ± 0.04	0.88 ± 0.03	**0.89 ± 0.03**
Recall	0.78 ± 0.04	**0.89 ± 0.03**	**0.89 ± 0.03**

The best metrics for each propulsion pattern are shown in bold

in pair-wise comparison of accuracies, F1-scores, precisions, and recalls obtained by these three classifiers (i.e., SVM with polynomial kernel, SVM with linear kernel, and MLP) ($p > 0.05$). Similarly, neither gender nor experience level significantly affected the observed accuracies, F1 scores, precisions, and recalls. Additionally, we observed No significant main effects of gender or experience, and no significant interaction with classifier ($p > 0.05$).

To visualize the performance variability and group-level comparisons, we generated boxplots of the evaluation metrics across all folds, stratified by classifier type, gender, and experience level (Fig. 5).

Discussion

We proposed a CWT-based method to transform the IMU's temporal data into a spectral format, revealing the intricate aspects of the propulsion mechanism. This method was chosen because propulsion cycles are non-stationary and their spectral content changes within each cycle. Our proposed method allowed us to identify different propulsion patterns during manual wheelchair propulsion. To our knowledge, only two prior studies have investigated the use of wearable measurements to classify propulsion patterns. Chen et al. [6] extracted time domain and frequency domain features directly from the IMU measurements. Time domain features included mean, standard deviation, and maximum value, while frequency domain features included zero crossing rate and autocorrelation. Chen et al. [6] focused on identifying the semicircular patterns from the other patterns and achieved 85% and 75% accuracy using a linear SVM model in indoor and outdoor manual wheelchair propulsion trials. Earlier, French et al. [12] demonstrated early feasibility with a wrist-mounted two-axis accelerometer in a single participant across multiple surfaces, achieving > 90% accuracy for four patterns using k-Nearest Neighbors and SVM. Taken together, these results are broadly consistent in magnitude with our findings, notwithstanding key differences in outcome definition (binary vs. four-class), protocol, and environment, which preclude any direct performance-level comparison. Previous studies have utilized wheel-mounted or body-worn sensors to monitor the physical activity of users [13–15, 21, 25, 28]. These studies mainly estimated high-level parameters that represent activity levels and types. Similarly, our method's performance agreed with previous studies in hand motion detection [11], which obtained

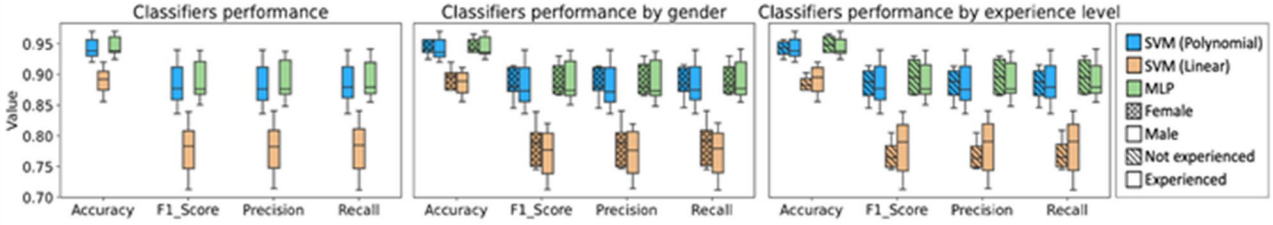


Fig. 5 Performance metrics for SVM with linear kernel, SVM with polynomial kernel, and MLP in identifying propulsion patterns and effect of gender and level of experience. Boxplots presented for results of 12-fold one-participant-out cross-validation approach

85% accuracy in detecting face-touch motions. Using CWT for feature extraction from the IMU measurements helped capture both time and frequency characteristics of non-stationary signals, providing better noise robustness and potentially improving the classification of patterns. The classifiers demonstrated consistent performance in identifying various propulsion patterns, as evidenced by a minimal standard deviation (SD ≤ 4%) across key metrics such as accuracy, F1-score, precision, and recall, consistently maintained across all propulsion patterns and classifiers (Table 4). Additionally, our method showed consistent performance among classifiers, gender, experience, and propulsion patterns, with no statistically significant differences detected (Friedman test, $p > 0.05$). Furthermore, previous studies have shown that the single-loop pattern is associated with higher cadence, reduced push percentage, and smaller contact angle compared to semicircular and double-loop patterns [2, 22, 23]. While the propulsion pattern itself may not directly cause injury, specific stroke-level characteristics—such as increased cadence, shorter push duration, smaller contact angle, and elevated peak push-rim force—are known to independently contribute to shoulder pain and pathology [2, 22, 23, 29, 33]. As such, automated pattern detection can serve as an indirect yet practical marker of cumulative exposure to these biomechanical risk factors in real-world settings. The ability of our method to accurately identify single-loop patterns is particularly important, as this pattern may signal elevated risk. Wheelchair users who predominantly employ this strategy may benefit from early preventive interventions aimed at reducing joint loading. By relating the propulsion patterns with the risk factors for shoulder pain in future work, we can provide actionable insights for manual wheelchair users and clinicians.

Despite these promising results, potential limitations need to be addressed in future studies. First, the dataset was collected using a roller platform, which, although beneficial for ensuring a controlled environment and consistent data collection, inherently restricts participants to straight-line propulsion. This may alter natural propulsion strategy compared with overground propulsion (turning, variable speed, uneven surfaces), reduce diversity of the recorded dataset, and affect generalizability of our proposed model. Thus, our setup may not fully replicate the variability and complexity of real-world wheelchair use, such as turning, maneuvering, or navigating uneven terrain. Additionally, participants were instructed to perform specific propulsion patterns, which, while necessary for training supervised algorithms, may deviate from their natural movement strategies. These factors may limit the generalizability of the trained models to spontaneous, unconstrained propulsion. Future work should validate the classifier in naturalistic settings, including overground propulsion and complex maneuvers, to ensure real-world applicability. Second, errors in segmenting the propulsion cycles could lead to incorrect feature extraction and classification. Third, the features extracted (Kurtosis, RMS, Maximum Value, Crest Factor, Shape Factor, Impulse Factor) might be redundant. Applying feature selection or dimensionality reduction techniques across various time and scale domains could enhance classifier performance, albeit at the expense of increased complexity. Therefore, it is crucial to ensure that this process enhances the representation of propulsion patterns without introducing unnecessary redundancy. We applied the Piecewise Cubic Hermite Interpolating Polynomial for time normalization to 2000 samples, followed by segmentation into five 400-sample segments. This approach is theoretically shape-preserving and minimizes overshoot and yielded high classification accuracy in our study. However, we did not specifically quantify its effect on the temporal dynamics of propulsion cycles. It is possible that minor distortions introduced during normalization and segmentation could limit the achievable accuracy, and future work should investigate this aspect in detail. Fourth, while the choice of SVM and MLP classifiers is justified to explore the feasibility of CWT, more complex models might be required for higher accuracy and robustness. Fifth, selecting 40 random recordings for each pattern per participant helps mitigate class imbalance. However, having more samples could further ensure that the selection process is truly unbiased and that the selected samples represent the entire dataset and wheelchair users. The small dataset size limited our ability to investigate feature selection and dimensionality reduction techniques. Addressing this limitation would necessitate the collection of a larger dataset.

Future research should investigate the performance of our proposed method in outdoor settings or indoor tracks and during complex wheelchair maneuvers such as turns or navigating an '8' shaped path. Additionally, enhancing this method's predictive capabilities by incorporating real-time analysis and feedback mechanisms could lead to a dynamic tool for both users and clinicians. These advancements would provide immediate insights into propulsion techniques and facilitate quick adjustments to prevent injury. The broader impact of our findings extends to enabling remote and long-term monitoring of propulsion patterns. Due to the compact and user-friendly nature of IMUs, clinicians and manual wheelchair users can leverage the insights provided by our method to monitor and adjust their propulsion patterns proactively. In this study, we mounted the IMU on the hand to maximize sensitivity to hand–rim interactions, which are critical for distinguishing propulsion patterns (e.g., semicircular vs. over-rim). While this

placement offered the highest discriminability for feasibility testing, the same algorithm could be deployed in practical formats such as gloves or low-profile straps, or adapted to wrist-worn devices after validation, enhancing its suitability for real-world use.

Conclusion

We demonstrated the feasibility of classifying manual wheelchair propulsion patterns from a hand-mounted IMU using a CWT-based pipeline with lightweight classifiers (SVM, MLP), achieving high accuracy across participants regardless of experience or gender. This shows that a simple, wearable system can capture cycle-level features linked to shoulder loading and injury risk. Moving beyond activity-level metrics, our method enables remote, long-term monitoring of stroke-level patterns in real-world use. These insights can support earlier, personalized interventions and training to reduce cumulative shoulder strain. Ultimately, this work advances accessible tools to promote healthier, more sustainable wheelchair propulsion with clear clinical and rehabilitation relevance.

Author contributions

R. F., A. K., N. R., H. R. and C. H. conceptualized the study and designed the experiments. R. F., A. K., H. R. and C. H. performed data collection and analysis. R. F., A. K., H. R. and C. H. contribute to the main manuscript text. R. F., A. K., H. R. and C. H. reviewed and approved the final version of the manuscript.

Funding

This work was financially supported by the Glenrose Rehabilitation Hospital, SMART Technology Innovation fund, and the University of Alberta Spinal Cord Injury Endowed Research Chair fund.

Data availability

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The study's experimental protocol received approval from the University of Alberta's Research Ethics Board Committee (No. Pro00115722), and all participants signed an informed consent form before participating in the study.

Consent for publication

Not applicable. This work does not contain any individual person's data in any form, including personal details, images, or videos.

Competing interests

The authors declare no competing interests.

Received: 25 October 2024 / Accepted: 24 November 2025
Published online: 10 December 2025

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